

Diodes

Introduction, Approximations, Series and parallel configuration, Problems

Gunn Diode



LED

PIN Diode



Step Recovery Diode



Laser Diode



Photo Diode



Schottky Diode

Tunnel Diode



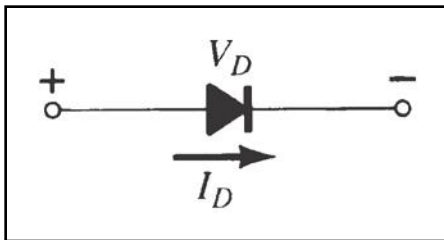
Varactor Diode



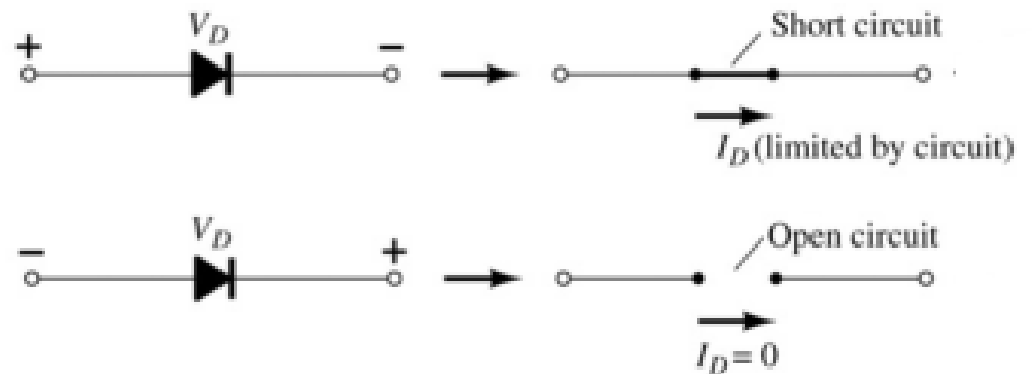
Zener Diode

Diodes

The diode is a 2-terminal device.



A diode ideally conducts in only one direction.

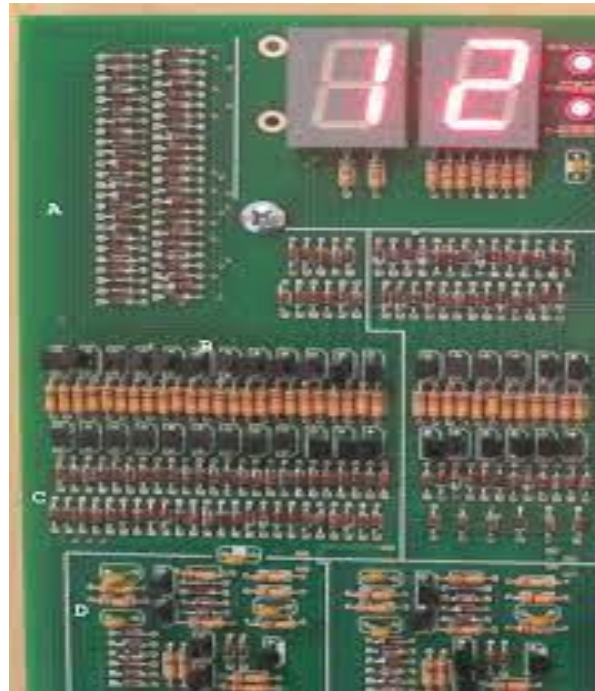


Application of diodes

► Some Common Applications of Diodes

- Rectifiers.
- Clipper Circuits.
- Clamping Circuits.
- Reverse Current Protection Circuits.
- In Logic Gates.
- Voltage Multipliers.

Are there diodes in computers



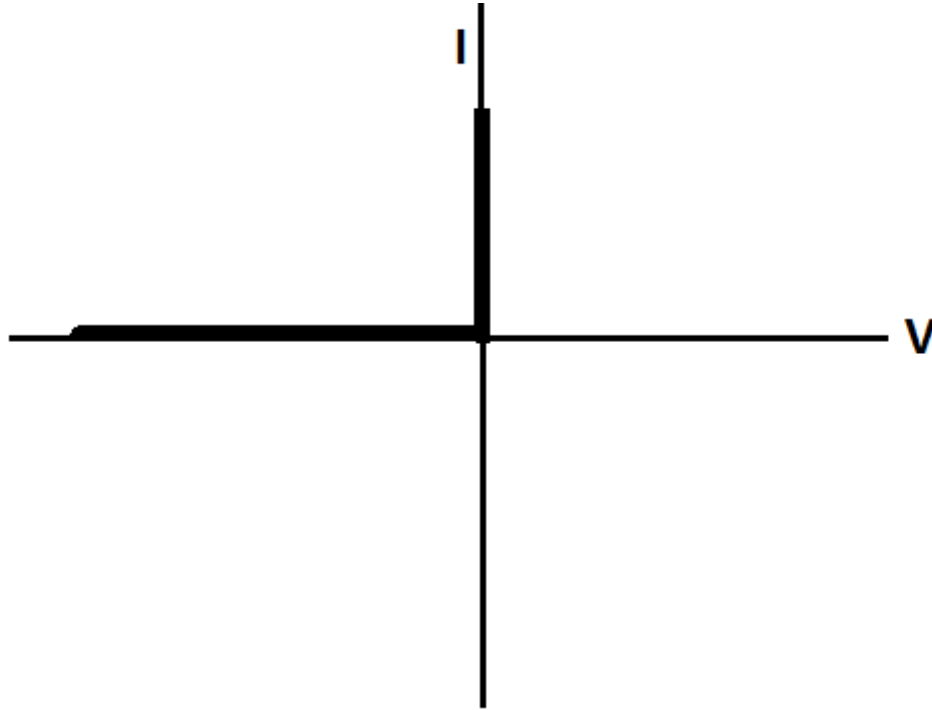
- ▶ Diode logic (DL), or diode-resistor logic (DRL), is the construction of Boolean logic gates from diodes. Diode logic was used extensively in the construction of early computers, where semiconductor diodes could replace bulky and costly active vacuum tube elements.

Importance in electronic Application

- ▶ The most common use of diodes is **to remove the negative component of an AC signal**. Because the negative portion of an AC waveform is usually identical to the positive half, very little information is effectively lost in this process of stripping it away, leading to more efficient signal processing

What is an Ideal Diode?

- ▶ An ideal diode is a diode that acts like a perfect conductor when voltage is applied forward biased and like a perfect insulator when voltage is applied reverse biased.
- ▶ So when positive voltage is applied across the anode to the cathode, the diode conducts forward current instantly. When voltage is applied in reverse, the diode conducts no current at all.

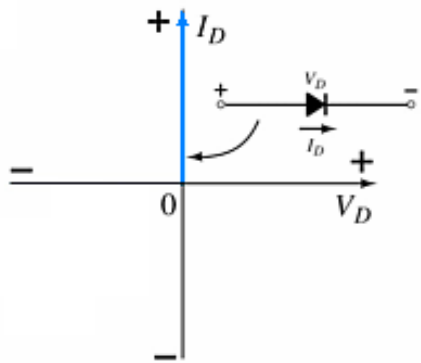


Ideal Diode Characteristics

curve when the diode receives forward voltage, it instantly conducts an infinite amount of current which it can supply to a circuit. When reverse voltage is fed to the diode, it conducts no current at all, no matter how great this reverse voltage is.

Diode Characteristics

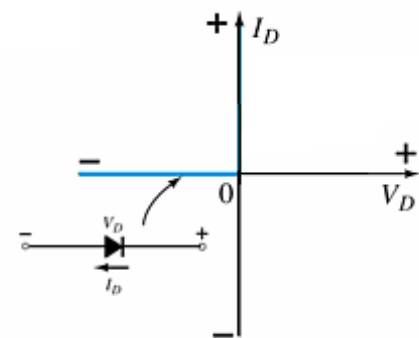
Conduction Region



The voltage across the diode is 0 V
The current is infinite
The forward resistance is defined as
$$R_F = V_F / I_F$$

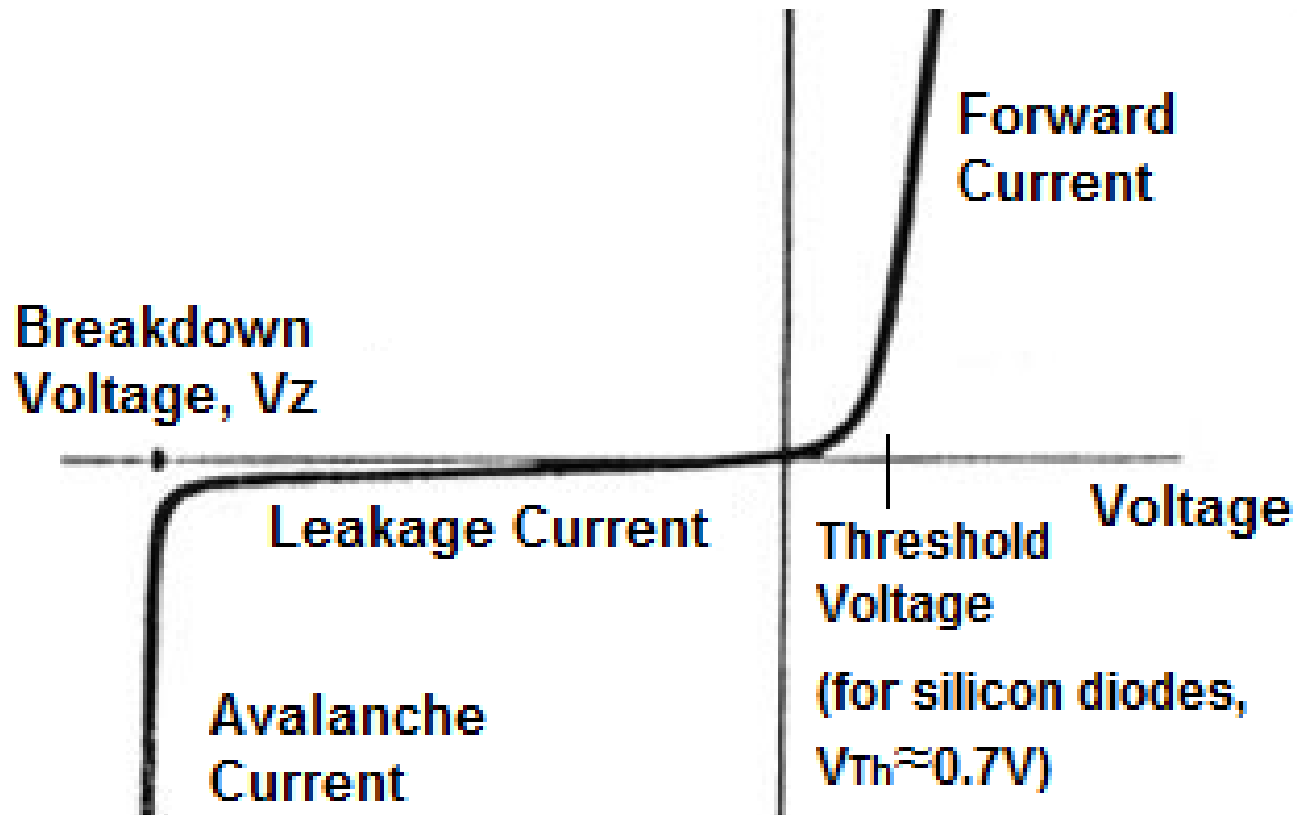
The diode acts like a short

Non-Conduction Region



All of the voltage is across the diode
The current is 0 A
The reverse resistance is defined as
$$R_R = V_R / I_R$$

The diode acts like open



Conventional Diode Characteristics Curve

Semiconductor Materials

Materials commonly used in the development of semiconductor devices:

Silicon (Si)

Germanium (Ge)

Gallium Arsenide (GaAs)

Doping

The electrical characteristics of silicon and germanium are improved by adding materials in a process called *doping*.

There are just two types of doped semiconductor materials:

n-type

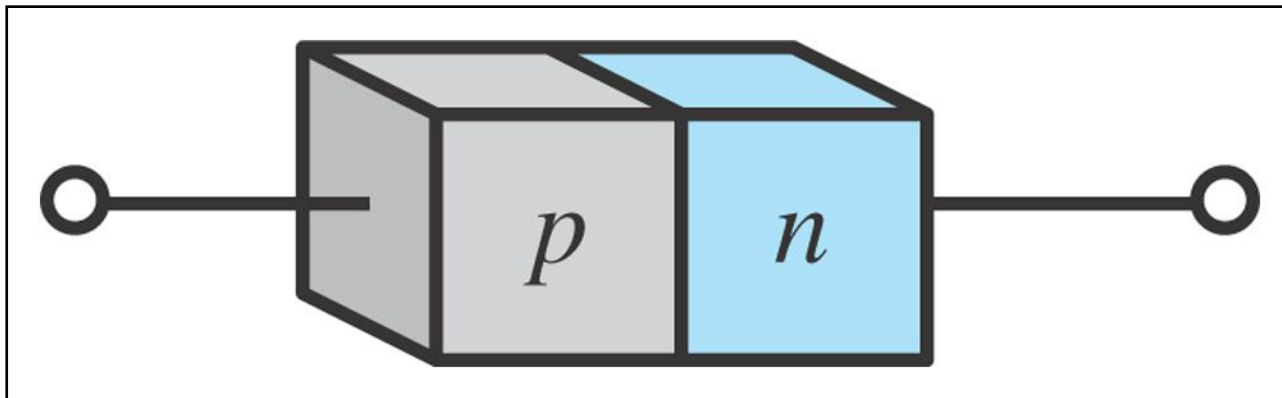
***n*-type** materials contain an excess of conduction band electrons.

p-type

***p*-type** materials contain an excess of valence band holes.

p-n Junctions

One end of a silicon or germanium crystal can be doped as a *p*-type material and the other end as an *n*-type material.



The result is a *p-n* junction

Pn junction

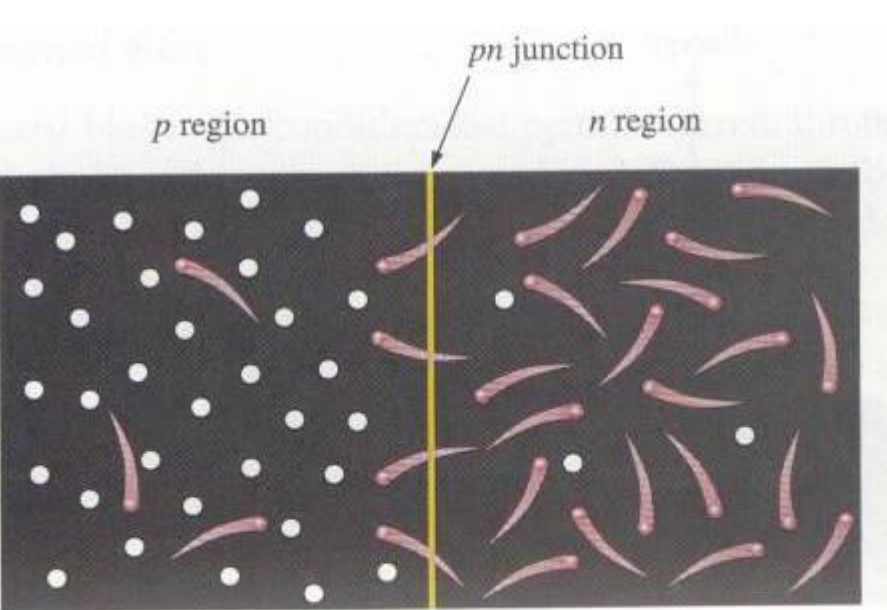
3 The PN-Junction

* The interface in-between p-type and n-type material is called a

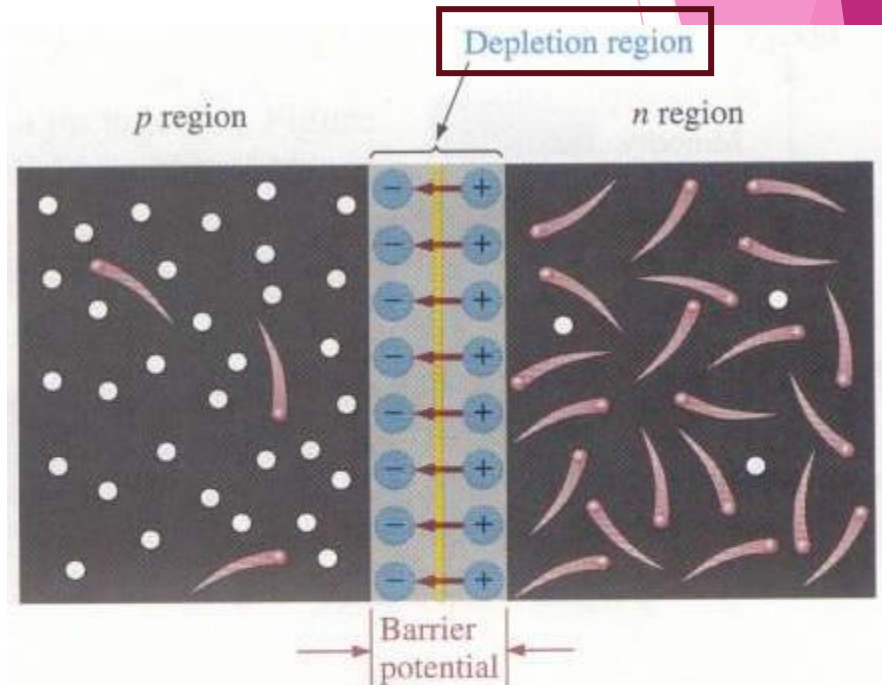
pn-junction.

The barrier potential $V_B \cong 0.6 - 0.7V$ for Si and $0.3V$ for Ge

at 300K: as $T \uparrow, V_B \downarrow$.



(a) At the instant of junction formation, free electrons in the n region near the pn junction begin to diffuse across the junction and fall into holes near the junction in the p region.

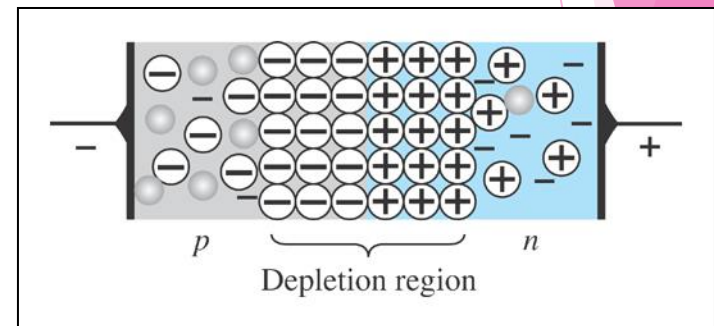


(b) For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the n region and a negative charge is created in the p region, forming a **barrier potential**. This action continues until the voltage of the barrier repels further diffusion.

p-n Junctions

At the *p-n* junction, the excess conduction-band electrons on the *n*-type side are attracted to the valence-band holes on the *p*-type side.

The electrons in the *n*-type material migrate across the junction to the *p*-type material (electron flow).



Electron migration results in a negative charge on the *p*-type side of the junction and a positive charge on the *n*-type side of the junction.

The result is the formation of a depletion region around the junction.

Diode Operating Conditions

A diode has three operating conditions:

No bias

Reverse bias

Forward bias

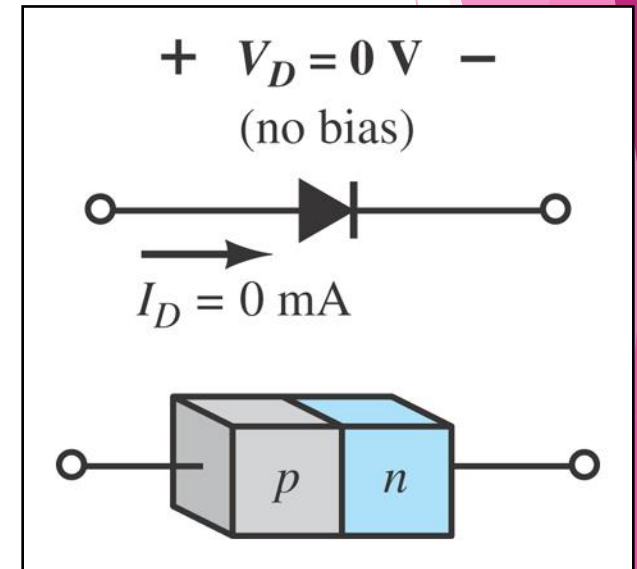
Diode Operating Conditions

No Bias

No external voltage is applied: $V_D = 0 \text{ V}$

There is no diode current: $I_D = 0 \text{ A}$

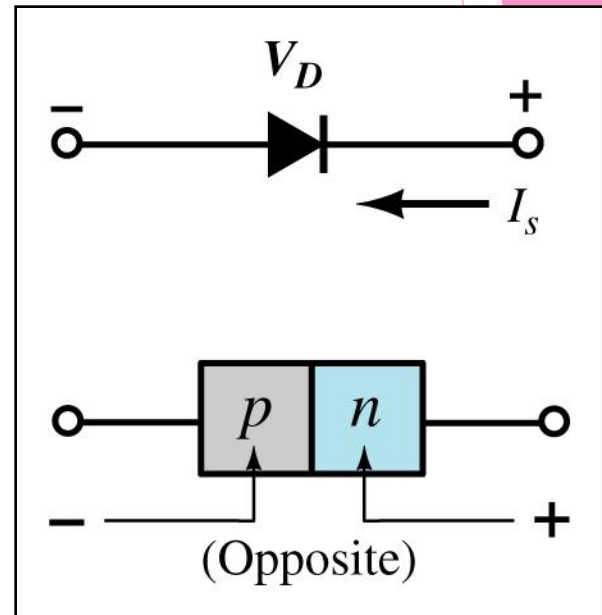
Only a modest depletion region exists



Diode Operating Conditions

Reverse Bias

External voltage is applied across the p - n junction in the opposite polarity of the p - and n -type materials.



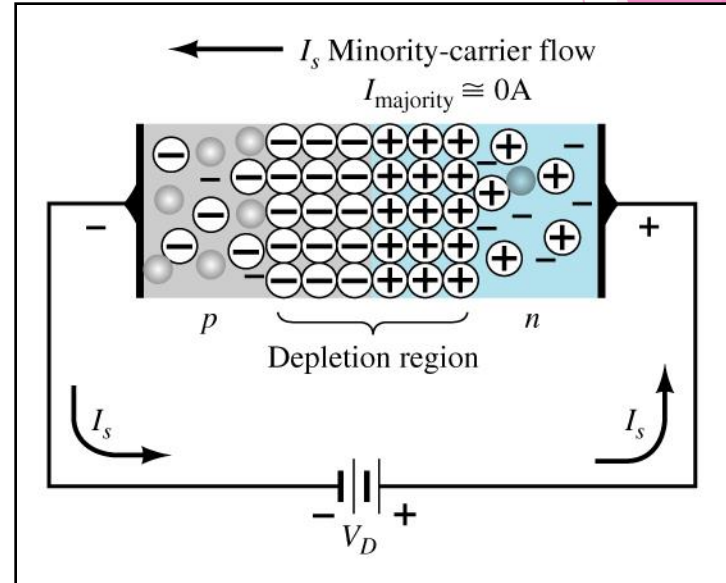
Diode Operating Conditions

Reverse Bias

The reverse voltage causes the depletion region to widen.

The electrons in the n -type material are attracted toward the positive terminal of the voltage source.

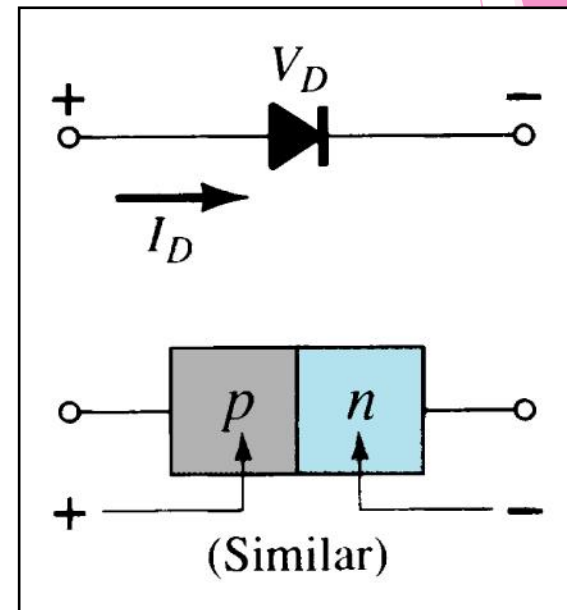
The holes in the p -type material are attracted toward the negative terminal of the voltage source.



Diode Operating Conditions

Forward Bias

External voltage is applied across the p - n junction in the same polarity as the p - and n -type materials.

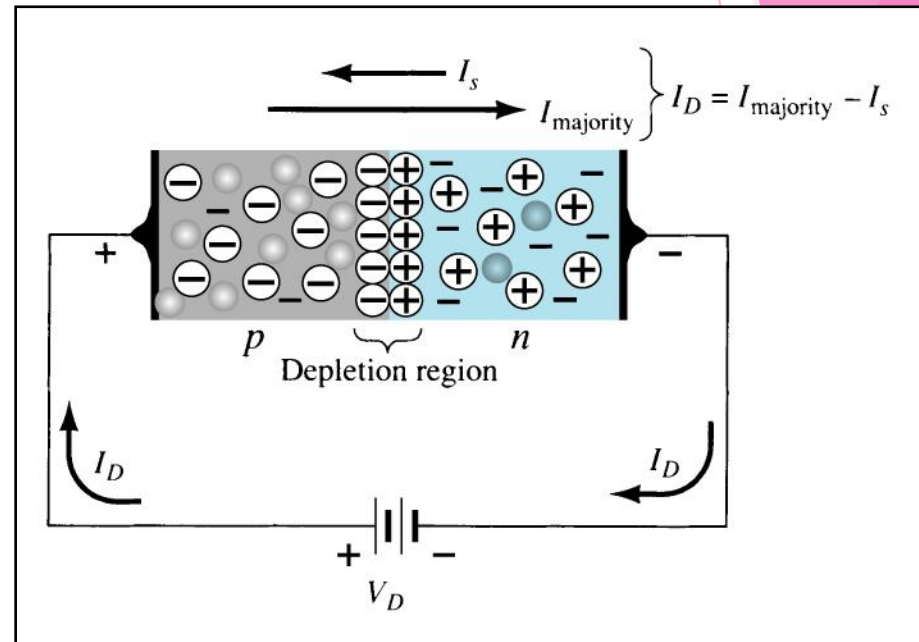


Diode Operating Conditions

Forward Bias

The forward voltage causes the depletion region to narrow.

The electrons and holes are pushed toward the p - n junction.



The electrons and holes have sufficient energy to cross the p - n junction.

Majority and Minority Carriers

Two currents through a diode:

Majority Carriers

The majority carriers in ***n*-type** materials are electrons.

The majority carriers in ***p*-type** materials are holes.

Minority Carriers

The minority carriers in ***n*-type** materials are holes.

The minority carriers in ***p*-type** materials are electrons.

Forward Bias Voltage

The point at which the diode changes from no-bias condition to forward-bias condition occurs when the electrons and holes are given sufficient energy to cross the p-n junction. This energy comes from the external voltage applied across the diode.

The forward bias voltage required for a:
**gallium arsenide diode $\cong 1.2 \text{ V}$ silicon diode $\cong$
 0.7 V , germanium diode $\cong 0.3 \text{ V}$**

Other Types of Diodes

There are several types of diodes besides the standard p - n junction diode. Three of the more common are:

Zener diodes

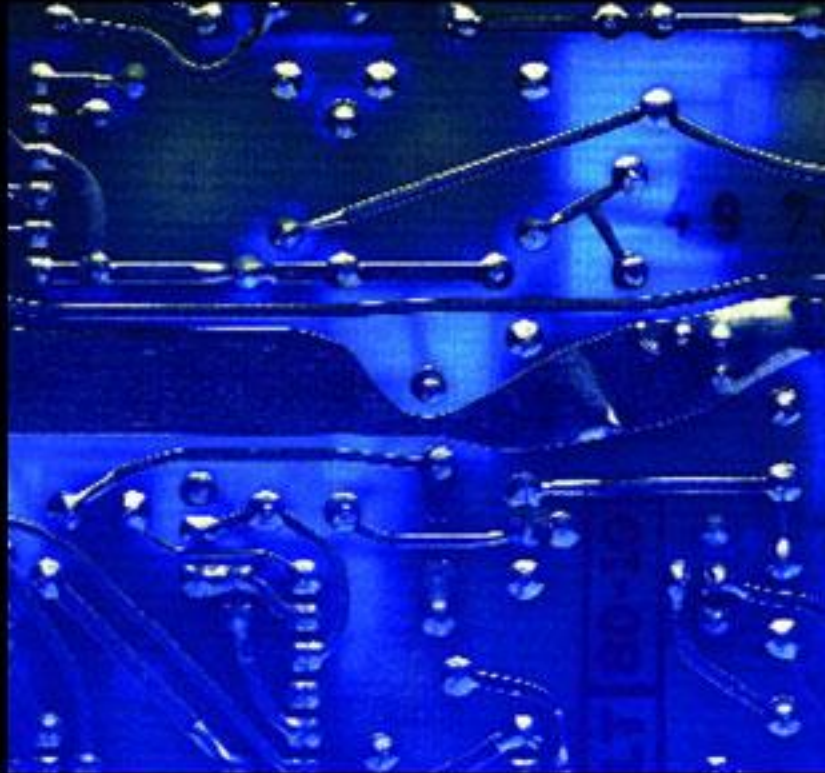
Light-emitting diodes

Diode arrays

ELECTRONIC DEVICES AND CIRCUIT THEORY

TENTH EDITION

BOYLESTAD



PEARSON

Chapter 2: Diode Applications

Series Diode Configurations

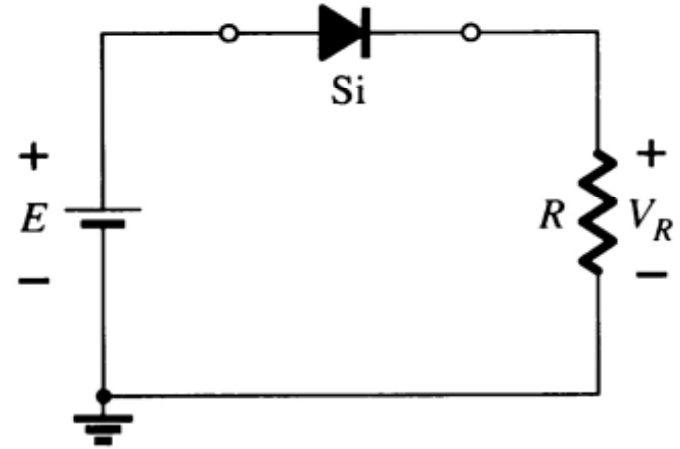
Forward Bias

Constants

- ▶ Silicon Diode: $V_D = 0.7 \text{ V}$
- ▶ Germanium Diode: $V_D = 0.3 \text{ V}$

Analysis (for silicon)

- ▶ $V_D = 0.7 \text{ V}$ (or $V_D = E$ if $E < 0.7 \text{ V}$)
- ▶ $V_R = E - V_D$
- ▶ $I_D = I_R = I_T = V_R / R$



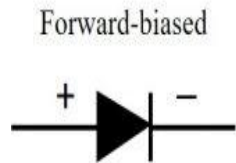
Diode Approximations

- ▶ **What is Diode Approximation?**
- ▶ Diode approximation is a mathematical method used to approximate the nonlinear behavior of real diodes to enable calculations and circuit analysis. There are three different approximations used to analyze the diode circuits.

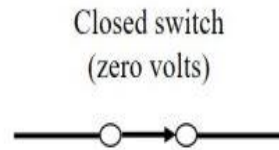
First diode approximation

- ▶ In the first approximation method, the diode is considered as a forward-biased diode and as a closed switch with zero voltage drop. It is not apt to use in real-life circumstances but used only for general approximations where preciseness is not required.

first-approximation

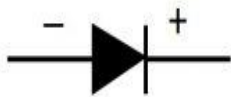


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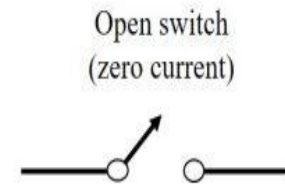


(a)

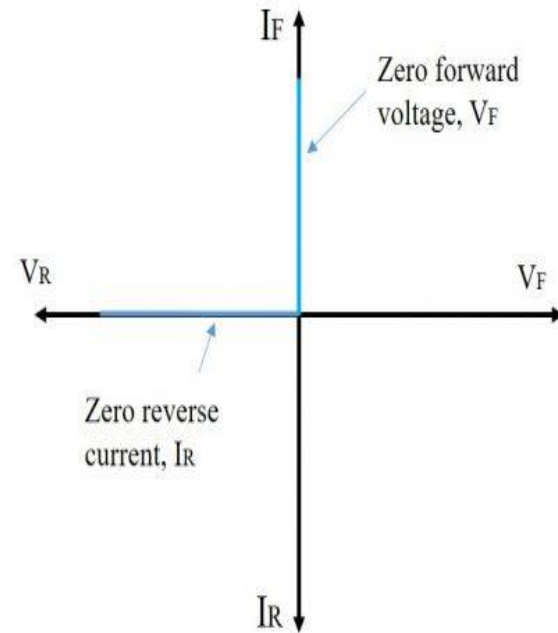
Reverse-biased



=



(b)



(c)

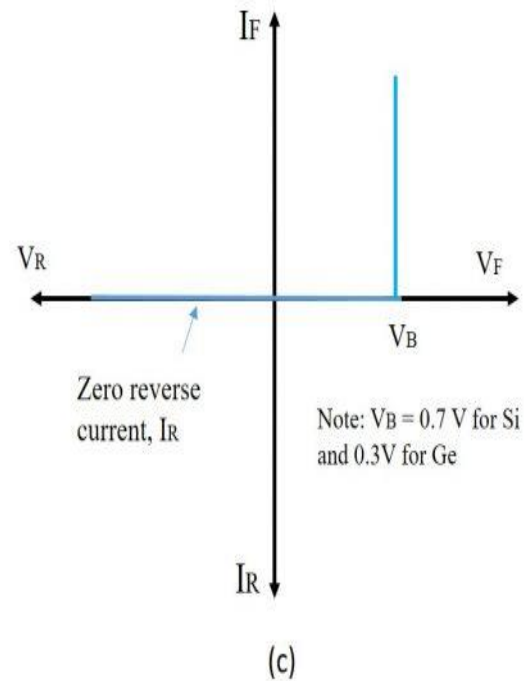
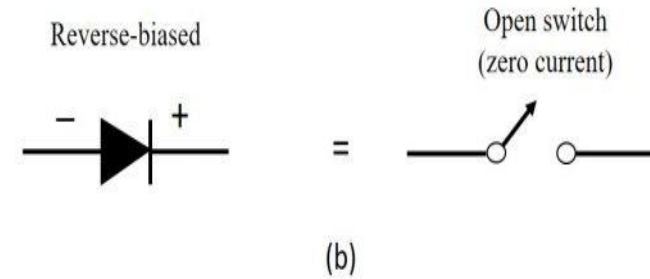
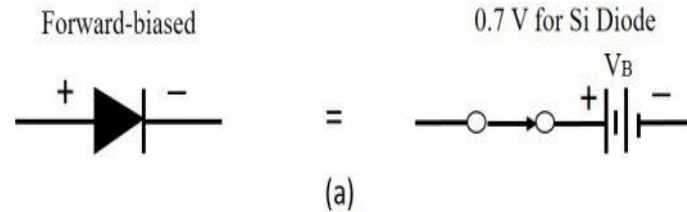
Ideal Diode

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Second diode Approximation

- ▶ In the second approximation, the diode is considered as a forward-biased diode in series with a battery to turn on the device. For a silicon diode to turn on, it needs 0.7V. A voltage of 0.7V or greater is fed to turn on the forward-biased diode. The diode turns off if the voltage is less than 0.7V.

Second Approximtion



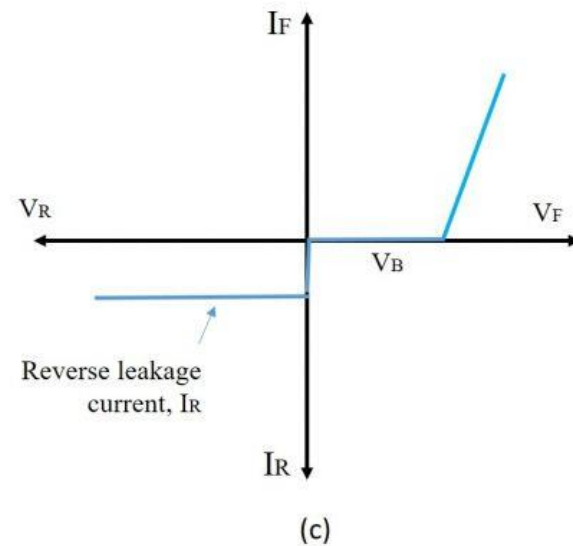
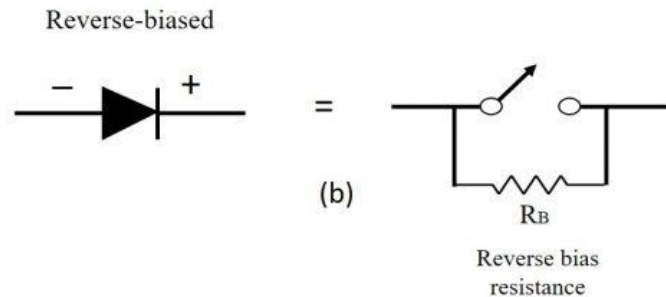
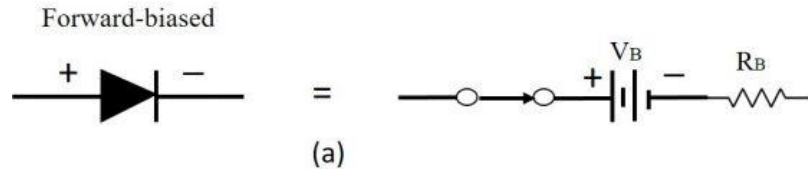
Third Approximation

- ▶ The third approximation of a diode includes voltage across the diode and voltage across bulk resistance, R_B . The bulk resistance is low, such as less than 1 ohm and always less than 10 ohms. The bulk resistance, R_B corresponds to the resistance of p and n materials. This resistance changes based on the amount of forwarding voltage and the current flowing through the diode at any given time.
- ▶ The voltage drop across the diode is calculated using the formula

Third Approximtion

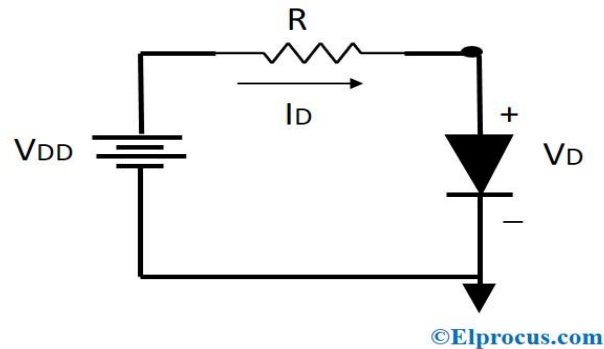
$$V_d = 0.7V + I_d * R_B$$

And if $R_B < 1/100 R_{Th}$ or $R_B < 0.001 R_{Th}$, we neglect that



Problem

- Use the second approximation of diode and find the current flowing through the diode.



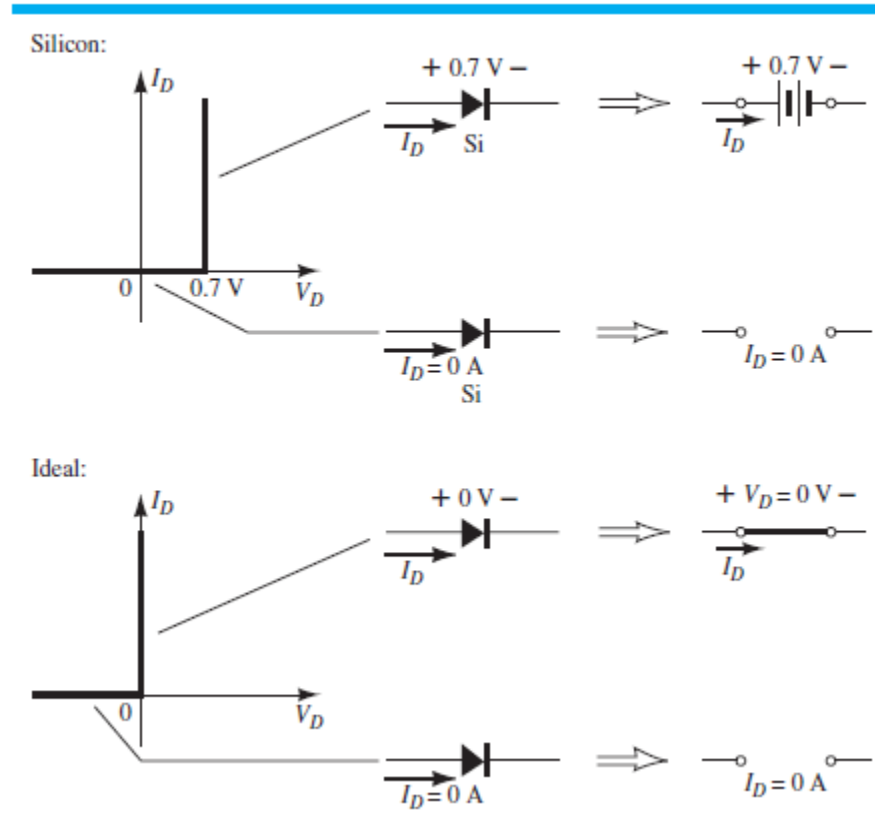
Solution

► $I_D = (V_s - V_D)/R = (4 - 0.7)/8 = 0.41A$

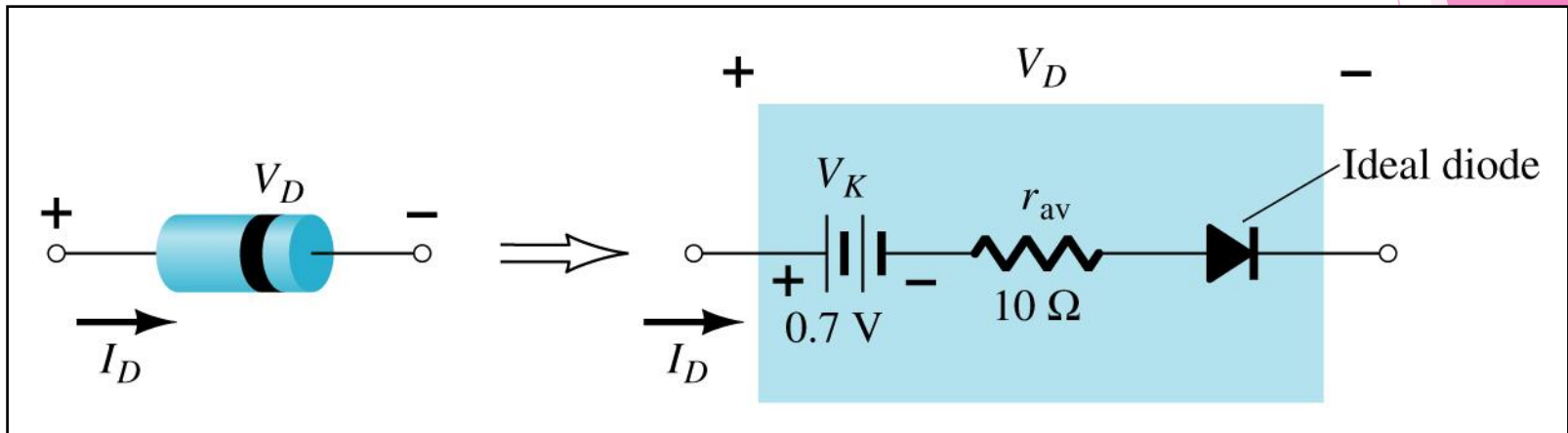
Series diode configuration.

TABLE 2.1

Approximate and Ideal Semiconductor Diode Models.



Diode Equivalent Circuit



Series diode configuration

In general, a diode is in the “on” state if the current established by the applied sources is such that its direction matches that of the arrow in the diode symbol, and $V_D \geq 0.7\text{ V}$ for silicon, $V_D \geq 0.3\text{ V}$ for germanium, and $V_D \geq 1.2\text{ V}$ for gallium arsenide.

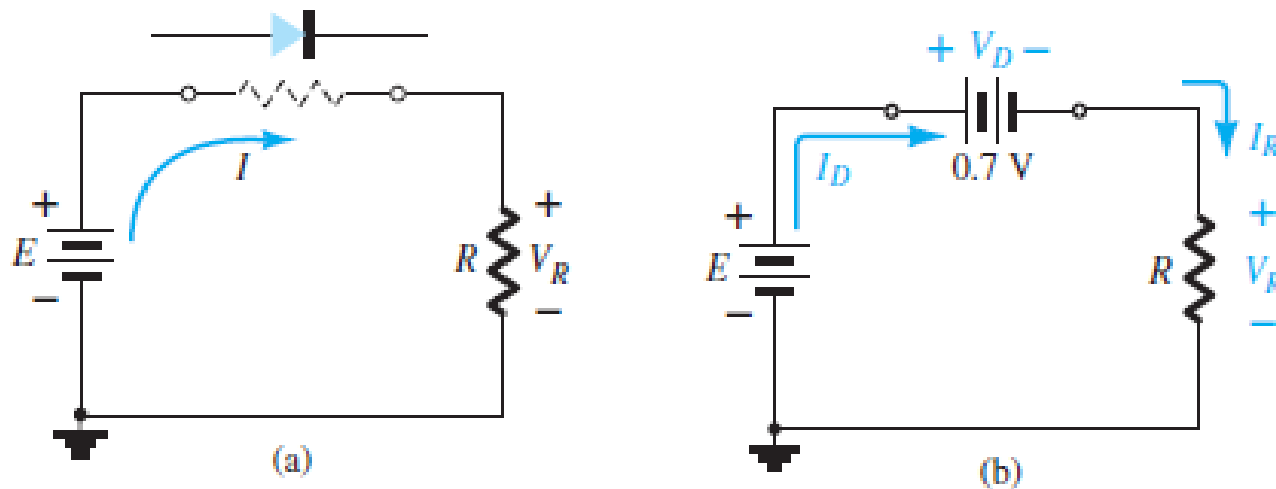


FIG. 2.9

(a) Determining the state of the diode of Fig. 2.8; (b) substituting the equivalent model for the “on” diode of Fig. 2.9a

Series diode configuration

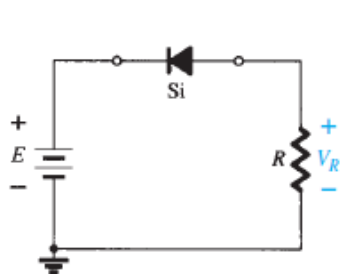


FIG. 2.10
Reversing the diode of Fig. 2.8.

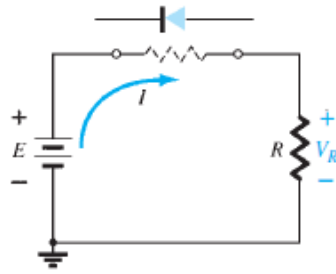


FIG. 2.11
Determining the state of the diode of Fig. 2.10.

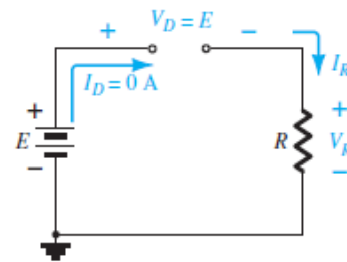


FIG. 2.12
Substituting the equivalent model for the "off" diode of Fig. 2.10.

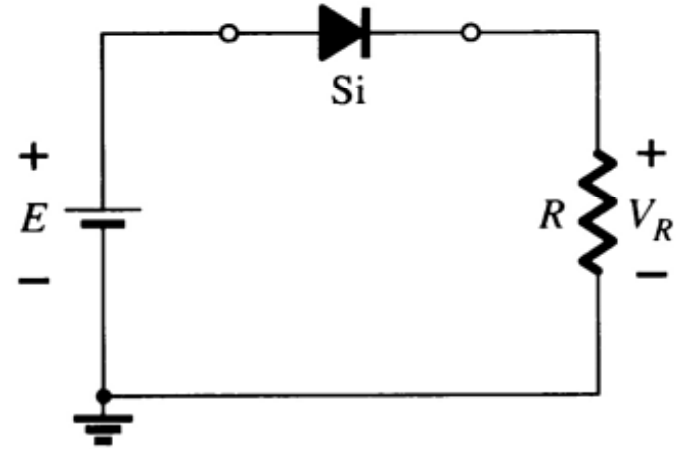
Series Diode Configurations

Reverse Bias

Diodes ideally behave as open circuits

Analysis

- ▶ $V_D = E$
- ▶ $V_R = 0 \text{ V}$
- ▶ $I_D = 0 \text{ A}$



*An open circuit can have any voltage across its terminals, but the current is always 0 A.
A short circuit has a 0-V drop across its terminals, but the current is limited only by the surrounding network.*

Series Diode Configurations

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D , V_R , and I_D .

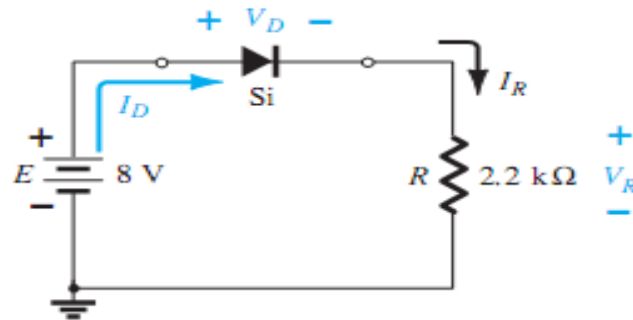


FIG. 2.13

Circuit for Example 2.4.

Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the “on” state,

$$V_D = 0.7\text{ V}$$

$$V_R = E - V_D = 8\text{ V} - 0.7\text{ V} = 7.3\text{ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\text{ V}}{2.2\text{ k}\Omega} \cong 3.32\text{ mA}$$

Series Diode Configurations

EXAMPLE 2.5 Repeat Example 2.4 with the diode reversed.

Solution: Removing the diode, we find that the direction of I is opposite to the arrow in the diode symbol and the diode equivalent is the open circuit no matter which model is employed. The result is the network of Fig. 2.14, where $I_D = 0$ A due to the open circuit. Since $V_R = I_R R$, we have $V_R = (0)R = 0$ V. Applying Kirchhoff's voltage law around the closed loop yields

$$E - V_D - V_R = 0$$

and

$$V_D = E - V_R = E - 0 = E = 8 \text{ V}$$

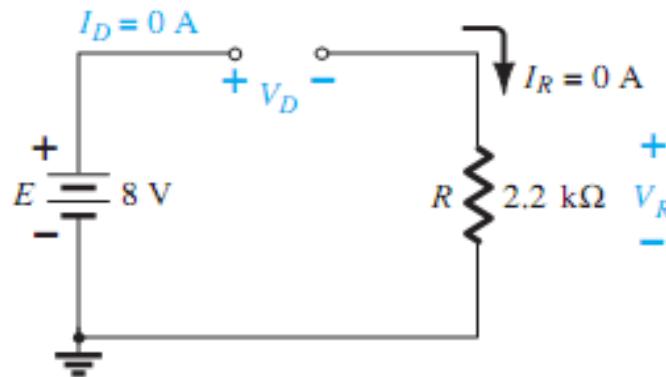


FIG. 2.14

*Determining the unknown quantities for
Example 2.5*

Series Diode Configurations

EXAMPLE 2.6 For the series diode configuration of Fig. 2.16, determine V_D , V_R , and I_D .

Solution: Although the “pressure” establishes a current with the same direction as the arrow symbol, the level of applied voltage is insufficient to turn the silicon diode “on.” The point of operation on the characteristics is shown in Fig. 2.17, establishing the open-circuit equivalent as the appropriate approximation, as shown in Fig. 2.18. The resulting voltage and current levels are therefore the following:

$$I_D = 0 \text{ A}$$

$$V_R = I_R R = I_D R = (0 \text{ A}) 1.2 \text{ k}\Omega = 0 \text{ V}$$

and

$$V_D = E = 0.5 \text{ V}$$

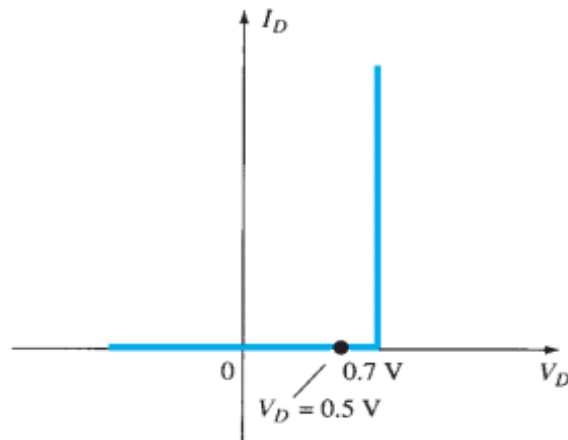


FIG. 2.17

Operating point with $E = 0.5 \text{ V}$.

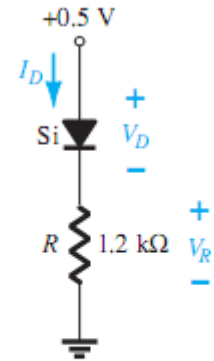


FIG. 2.16

Series diode circuit for Example 2.6.

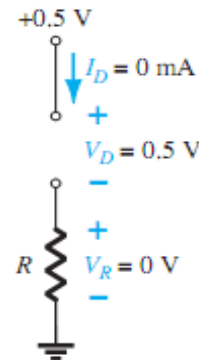


FIG. 2.18

Determining I_D , V_R , and V_D for the circuit of Fig. 2.16.

Series Diode Configurations

EXAMPLE 2.7 Determine V_o and I_D for the series circuit of Fig. 2.19.

Solution: An attack similar to that applied in Example 2.4 will reveal that the resulting current has the same direction as the arrowheads of the symbols of both diodes, and the network of Fig. 2.20 results because $E = 12\text{ V} > (0.7\text{ V} + 1.8\text{ V [Table 1.8]}) = 2.5\text{ V}$. Note the redrawn supply of 12 V and the polarity of V_o across the $680\text{-}\Omega$ resistor. The resulting voltage is

$$V_o = E - V_{K_1} - V_{K_2} = 12\text{ V} - 2.5\text{ V} = 9.5\text{ V}$$

and

$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{680\text{ }\Omega} = 13.97\text{ mA}$$

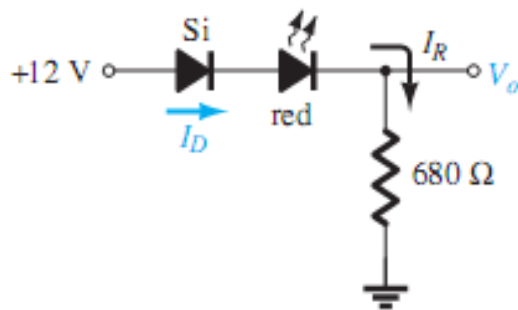


FIG. 2.19

Circuit for Example 2.7.

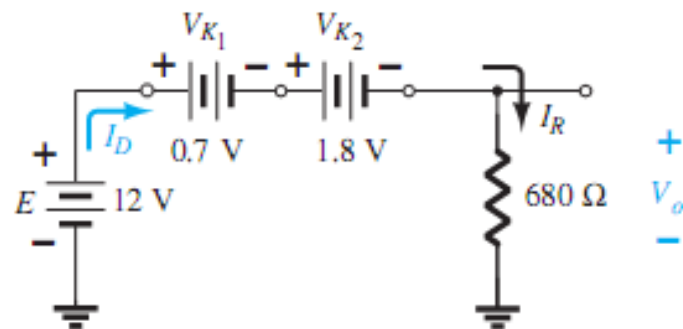
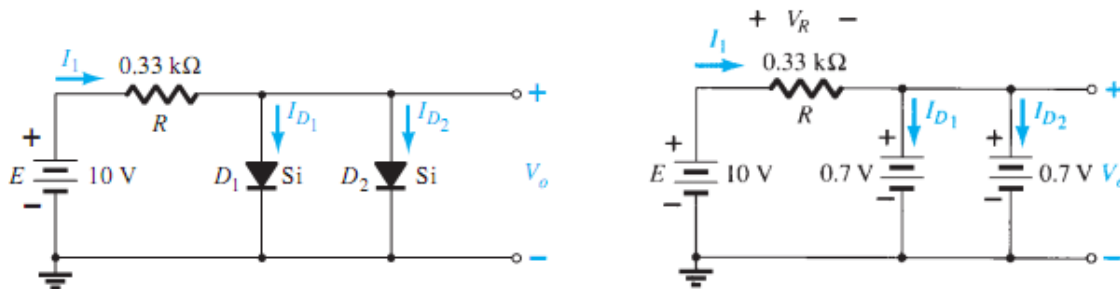


FIG. 2.20

Determining the unknown quantities for Example 2.7.

Parallel Configurations

EXAMPLE 2.10 Determine V_o , I_1 , I_{D1} , and I_{D2} for the parallel diode configuration of Fig. 2.28.



Solution: For the applied voltage the “pressure” of the source acts to establish a current through each diode in the same direction as shown in Fig. 2.29. Since the resulting current direction matches that of the arrow in each diode symbol and the applied voltage is greater than 0.7 V, both diodes are in the “on” state. The voltage across parallel elements is always the same and

$$V_o = 0.7 \text{ V}$$

The current is

$$I_1 = \frac{V_R}{R} = \frac{E - V_D}{R} = \frac{10 \text{ V} - 0.7 \text{ V}}{0.33 \text{ k}\Omega} = 28.18 \text{ mA}$$

Assuming diodes of similar characteristics, we have

$$I_{D1} = I_{D2} = \frac{I_1}{2} = \frac{28.18 \text{ mA}}{2} = 14.09 \text{ mA}$$

EXAMPLE 2.13 Determine the currents I_1 , I_2 , and I_{D_2} for the network of Fig. 2.37.

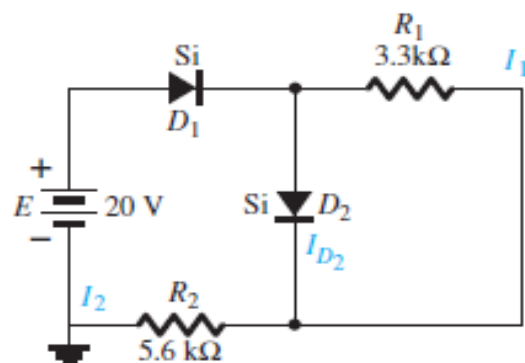


FIG. 2.37

Network for Example 2.13.

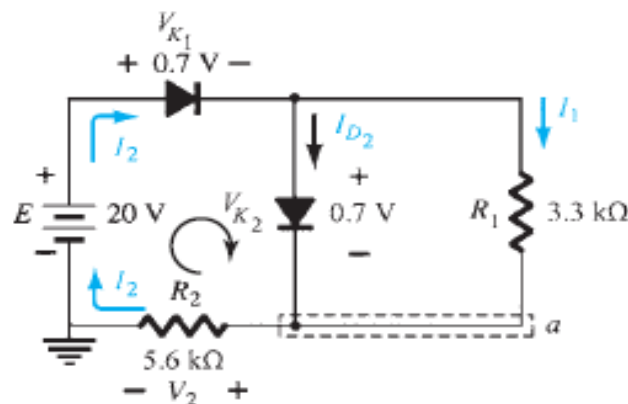


FIG. 2.38

Determining the unknown quantities for Example 2.13.

Solution: The applied voltage (pressure) is such as to turn both diodes on, as indicated by the resulting current directions in the network of Fig. 2.38. Note the use of the abbreviated notation for “on” diodes and that the solution is obtained through an application of techniques applied to dc series-parallel networks. We have

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7 \text{ V}}{3.3 \text{ k}\Omega} = 0.212 \text{ mA}$$

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

and
$$V_2 = E - V_{K_1} - V_{K_2} = 20 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = 18.6 \text{ V}$$

with
$$I_2 = \frac{V_2}{R_2} = \frac{18.6 \text{ V}}{5.6 \text{ k}\Omega} = 3.32 \text{ mA}$$

At the bottom node a ,

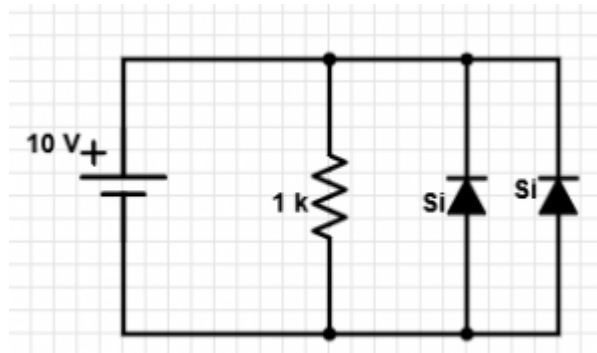
$$I_{D_2} + I_1 = I_2$$

and
$$I_{D_2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} \cong 3.11 \text{ mA}$$

Quick check 1

From the given circuit, what is the value of current flowing through the 1 k resistor parallel to the diodes?

- a) 10 mA
- b) 9.3 mA
- c) 8.6 mA
- d) 0 mA

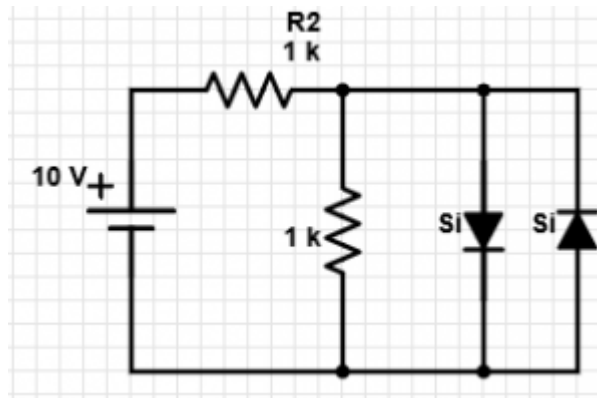


Quick check 1

- ▶ Answer: a
Explanation: As both the diodes are reverse biased.
Voltage drop across the resistor = 10 V. Hence, current
 $= 10 \text{ V} / 1 \text{ k} = 10 \text{ mA}$.

Quick check 2

From the given circuit, what is the value of current flowing through the 1 k resistor?



- a) 0 mA
- b) 10 mA
- c) 9.3 mA
- d) 0.7 mA

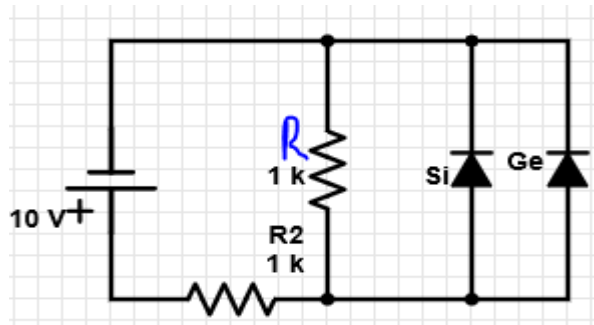
Quick check 2

- ▶ Answer: d
Explanation: As one of the diodes is forward biased, voltage drop across it = 0.7 V. Now, as this diode is in parallel with the given resistor, voltage across resistor = 0.7 V $\Rightarrow$ current = 0.7 mA.

Quick check 3

In the given circuit, what is the value of current through R?

- a) 0.7 mA
- b) 0.3 mA
- c) 1 mA
- d) 10 mA



Quick check 3

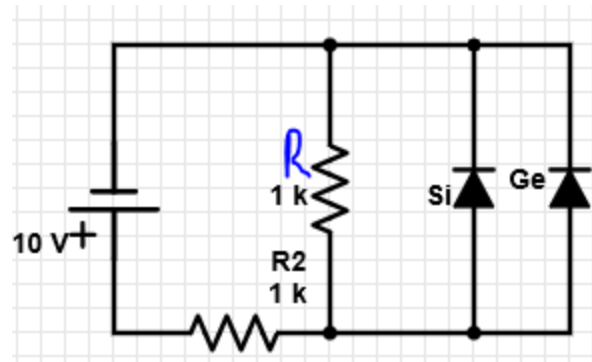
► Answer: b

Explanation: The source voltage initially increases from 0-10 V. As the source voltage increases, the voltage across the diodes also increases. When the voltage across the diodes reaches a value of 0.3 V, the Germanium diode starts conducting, whereas the silicon diode is still in the off state. Hence, the voltage across R = 0.3 V.

Quick check 4

. In the given circuit, what is the value of current through the germanium diode?

- a) 9.4 mA
- b) 9.7 mA
- c) 0.3 mA
- d) 0 mA



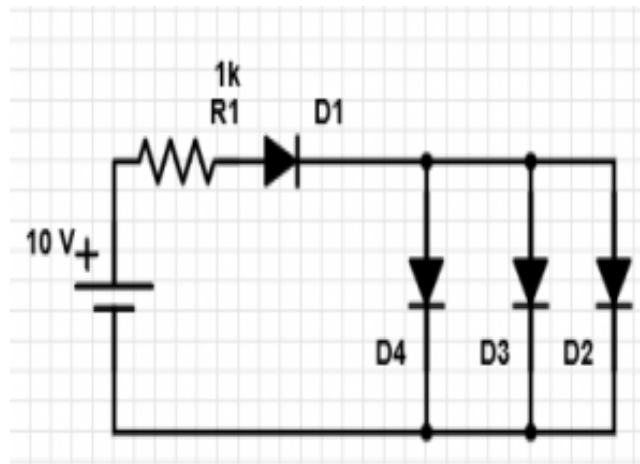
Quick check 4

- ▶ Answer: a
Explanation: Current through $R_2 = 9.7 \text{ mA}$. Also, current through $R = 0.3 \text{ mA}$, current through the silicon diode $= 0 \text{ mA}$. Hence, from the KCL, we get the current through the germanium diode to be 9.4 mA .

Quick check 5

In the given circuit, if the diode D2 is a Germanium diode and all other diodes are Silicon diodes then which of the following statements is true?

- a) $I_{D2} < I_{D3}; I_{D2} = I_{D1}$
- b) $I_{D2} > I_{D3}; I_{D2} > I_{D1}$
- c) $I_{D2} > I_{D3}; I_{D2} = I_{D1}$
- d) $I_{D2} > I_{D3}; I_{D2} < I_{D1}$



Quick check 5

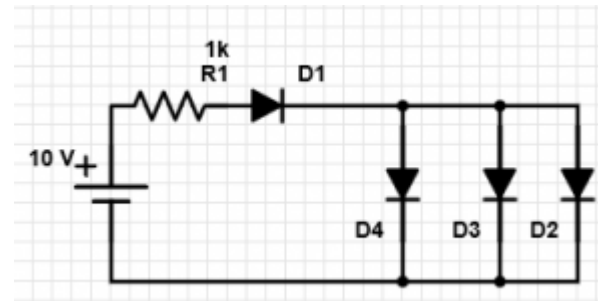
► Answer: c

Explanation: Voltage across the parallel diodes = 0.3 V. Hence, current through D3 = 0. Hence $I_{D2} > I_{D3}$. Also, as both the parallel silicon diodes are in parallel and in off-state, the diodes 1 and 2 are in series and hence, current through them is same.

Quick check 6

In the circuit, considering the diode 1,4 to be a Germanium diode and the diodes 2,3 to be silicon diodes, which of the following statements are true?

- a) $I_{D1} = I_{D2}$
- b) $I_{D1} = I_{D3}$
- c) $I_{D1} = I_{D4}$
- d) $I_{D1} > I_{D4}$



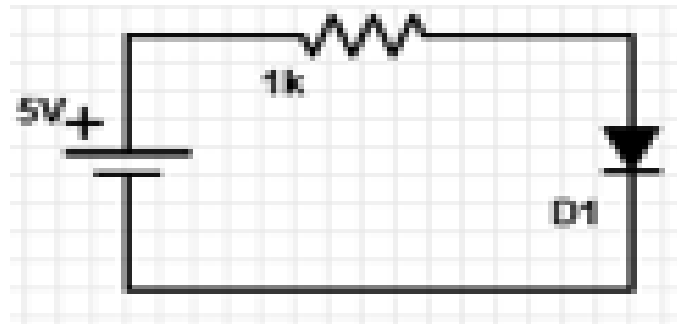
Quick check 6

► Answer: c

Explanation: In the given circuit, the silicon diodes are in off-state as the voltage across them is 0.3 V. Hence, they act as open circuit and the diodes 1 and 4 are in series and hence the current through diodes 1 and 4 is same.

Quick check 7

Assuming the diode in the given circuit diagram to be a silicon p-n junction diode, what is the current for the given circuit diagram?



- a) 0
- b) 5 mA
- c) 4.3 mA
- d) Can't be determined

Quick check 7

► Answer: c

Explanation: 4.3 mA. The silicon diode in the circuit is forward biased. Considering a voltage drop of 0.7 V across the diode, the voltage drop across the resistor is $(5 - 0.7) \text{ V} = 4.3 \text{ V}$. Hence, Current in the circuit = Current through the resistor = $(4.3/1000) \text{ A} = 4.3 \text{ mA}$.